

MATH GU4053 TOPOLOGY
HOMEWORK 1

DUE: WEDNESDAY, FEBRUARY 4, 2026, BEFORE 23:59PM

SUBMIT THE COMPLETED ASSIGNMENT TO <https://www.gradescope.com>

- This assignment has **5 questions** on **two pages**.
 - This assignment counts towards 2% of your final grade.
 - Your solutions must be *handwritten*. For instance on a tablet/iPad, or with pen and paper scanned to a pdf.
 - While collaborative work is allowed and generally beneficial, you must write up the solutions *on your own* and *in your own words*.
 - Rigor, clarity, and proper use of mathematical notation count! The work you submit for evaluation will require showing all steps and a clear explanation of what you have done. The explanations need not be lengthy to be clear.
 - Solutions based on an advanced result which has not been covered in this class, or one of its prerequisites, will not receive full credit. (Especially if the use of an advanced result significantly changes the difficulty of the problem.)
-

1. Let X be an indiscrete topological space (i.e., the only open subsets are $\emptyset$ and X). Prove that X is contractible if and only if $X \neq \emptyset$.

Your proof should treat “if” and “only if” separately.

2. Let X be any topological space and let Y be a discrete topological space (i.e., all subsets of Y are open).
- (i) Prove that any homotopy

$$F : I \times X \rightarrow Y$$

is a constant homotopy (i.e., that $F(t, x) = F(0, x)$ for any $t \in I$ and any $x \in X$). *What property of I does your proof use?*

- (ii) Deduce that two maps $f_0, f_1 : X \rightarrow Y$ are homotopic if and only if $f_0 = f_1$.

- (iii) Deduce that two discrete topological spaces X and X' are homotopy equivalent if and only if there exists a bijection $X \rightarrow X'$.

Your proof should treat “if” and “only if” separately.

3. Let

$$X = \{0\} \cup \left\{\frac{1}{n} \mid n \in \mathbb{Z}_{>0}\right\},$$

$$Y = \{-1\} \cup \left\{\frac{1}{n} \mid n \in \mathbb{Z}_{>0}\right\},$$

both equipped with the subspace topology from $\mathbb{R}$.

- (i) Explain why $\{y\} \subset Y$ is open for any $y \in Y$, and deduce that the topology on Y is discrete.
- (ii) Show that X and Y are not homotopy equivalent.

Make sure your proof in (ii) is rigorous. One strategy is an indirect proof: assume for contradiction that $f : X \rightarrow Y$ is a homotopy equivalence with homotopy inverse g , and use the previous exercise to deduce that f is surjective. Then prove that any continuous map $X \rightarrow Y$ has finite image, and deduce a contradiction. If you use that X is compact, you should explain why.

4. For a topological space X , let

$$\text{Conf}_2(X) = \{(x_1, x_2) \in X \times X \mid x_1 \neq x_2\}.$$

- (i) Show that

$$\text{Conf}_2(\mathbb{R}^n) \rightarrow S^{n-1}$$

$$(x_1, x_2) \mapsto \frac{x_2 - x_1}{|x_2 - x_1|}$$

is a homotopy equivalence for any n .

Your proof should involve an explicit homotopy inverse, and explanation for the relevant homotopies.

- (ii) True or false: if $X \simeq Y$ then $\text{Conf}_2(X) \simeq \text{Conf}_2(Y)$?
Provide either a proof, or a counter-example with justification.

5. A topological space X is called *weakly contractible* if for any integer $n \geq 0$ and any continuous map $f : S^{n-1} \rightarrow X$, there exists a map $F : D^n \rightarrow X$ with $F|_{S^{n-1}} = f$.

- (i) Prove that a weakly contractible space is non-empty and path connected.
- (ii) Prove that a contractible space is weakly contractible.

Hint for part (i): The definitions $D^n = \{x \in \mathbb{R}^n \mid |x| \leq 1\}$ and $S^{n-1} = \{x \in \mathbb{R}^n \mid |x| = 1\}$ make sense also for $n = 0$. It is not a typo that this case is included!